

LOS ANGELES HAWTHORNE, CA, USA

WMO: 722956

Lat: 33.923N

Lon: 118.334W

Elev: 63

StdP: 14.66

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 03167

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 44.5	(c) 45.9	(d) 11.5	(e) 9.9	(f) 65.6	(g) 17.1	(h) 13.1	(i) 64.1	(j) 17.9	(k) 58.7	(l) 15.4	(m) 59.6	(n) 1.0	(o) 70	(p) 0.348

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	88.4	63.5	84.6	63.6	81.6	63.9	70.7	80.1	69.4	78.3	68.4	77.0	6.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.3	100.7	75.5	66.0	96.1	74.7	64.7	91.7	73.0	34.6	80.1	33.5	78.7	32.6	77.0	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.3	(b) 14.0	(c) 12.5	(e) 39.4	(f) 97.2	(g) 2.8	(h) 5.0	(i) 37.4	(j) 100.8	(k) 35.7	(l) 103.7	(m) 34.1	(n) 106.5	(o) 32.1	(p) 110.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.7	58.8	58.6	60.7	62.2	64.5	67.2	70.9	71.5	71.2	68.4	63.5	58.1
	DBStd	6.62	5.11	5.00	4.70	4.77	3.81	3.00	3.65	3.51	4.53	5.30	5.18	4.36
	HDD50	2	1	0	0	0	0	0	0	0	0	0	0	1
	HDD65	1039	202	189	152	111	51	8	1	0	1	19	87	218
	CDD50	5352	275	241	331	365	449	517	648	666	635	570	404	251
	CDD65	913	10	10	18	26	36	75	184	201	185	124	41	3
	CDH74	4013	115	103	141	197	182	150	563	662	820	715	311	54
(13) CDH80		1058	18	25	38	66	66	29	89	111	240	272	100	4
(14) Wind	WSAvg	4.7	3.1	4.2	5.0	5.8	6.0	5.8	5.8	5.5	5.0	4.0	3.2	3.0
(15) Precipitation	PrecAvg	12.1	2.6	2.8	1.8	0.7	0.1	0.1	0.0	0.0	0.2	0.4	1.3	1.9
	PrecMax	29.5	12.7	13.3	6.4	4.5	0.9	1.0	0.4	1.3	1.9	3.8	7.5	8.8
	PrecMin	3.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.6	2.8	3.0	1.7	0.9	0.2	0.2	0.1	0.2	0.4	0.6	1.5	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.3	84.0	84.4	88.7	88.4	84.0	86.8	87.2	93.0	94.1	89.1	79.7
		MCWB	55.2	56.6	58.0	59.5	60.1	66.9	70.1	69.2	68.1	62.5	59.3	54.1
	2%	DB	77.0	76.5	77.7	79.2	78.7	77.7	82.1	83.1	86.2	87.1	82.2	74.1
		MCWB	53.0	54.1	55.6	57.1	61.0	65.4	68.8	68.4	67.3	60.9	57.6	53.3
	5%	DB	72.7	71.3	72.4	73.4	73.1	74.7	79.4	80.2	81.9	81.6	76.7	70.2
		MCWB	52.8	54.1	55.5	57.1	61.2	64.4	68.2	67.8	67.2	61.5	55.8	52.5
	10%	DB	68.0	66.3	68.2	69.6	70.2	72.6	77.0	77.5	78.5	76.7	72.1	65.9
		MCWB	52.6	54.1	55.8	56.8	60.0	63.3	67.2	67.1	66.4	62.1	56.2	52.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.5	61.0	62.1	62.8	66.1	68.4	72.3	71.6	72.3	69.7	63.8	61.1
		MCDB	66.7	68.8	73.8	76.0	78.9	79.6	82.7	81.3	84.7	79.1	73.4	65.3
	2%	WB	58.9	59.1	60.2	60.5	63.7	66.4	70.3	70.2	70.4	67.5	62.2	59.3
		MCDB	64.4	65.8	69.3	71.5	72.9	75.6	79.1	79.4	80.4	77.5	70.3	64.5
	5%	WB	57.5	57.7	58.9	59.3	62.2	65.4	69.1	69.0	69.0	65.6	60.9	57.8
		MCDB	63.9	64.4	66.8	68.7	70.3	73.5	77.2	77.8	78.3	74.5	68.5	63.2
	10%	WB	56.3	56.5	57.7	58.2	60.9	64.3	68.0	67.9	67.6	64.3	59.7	56.5
		MCDB	62.8	63.3	64.5	66.3	68.2	71.1	75.6	76.3	76.4	72.2	67.2	62.1
(35) Mean Daily Temperature Range	5% DB	MDBR	17.2	15.5	14.1	13.3	11.2	10.5	11.4	11.7	12.7	14.6	16.8	16.7
		MCDBR	25.7	24.5	22.1	21.5	17.5	13.6	14.0	14.5	18.6	24.4	26.0	24.3
		MCWBR	11.0	10.6	9.1	8.6	6.9	5.7	5.5	5.8	6.9	9.3	11.4	12.0
	5% WB	MCDBR	16.2	16.8	15.4	15.8	13.9	12.9	13.2	13.3	15.5	16.7	17.5	15.7
		MCWBR	8.4	8.5	7.0	7.0	6.1	5.6	5.3	5.6	6.1	7.0	9.0	9.0
(40) Clear-Sky Solar Irradiance	taub		0.339	0.353	0.377	0.379	0.383	0.399	0.399	0.381	0.367	0.374	0.323	0.325
	taud		2.488	2.461	2.366	2.372	2.392	2.348	2.415	2.448	2.480	2.430	2.574	2.570
	Ebn at Noon		276	284	284	288	286	280	279	284	284	272	281	276
	Edh at Noon		27	31	37	38	38	40	37	35	33	32	25	24
(44) All-Sky Solar Radiation	RadAvg		1000	1282	1714	2098	2261	2353	2354	2240	1892	1437	1112	907
	RadStd		72	127	97	109	156	146	128	76	80	108	69	69

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.37	N/A	-2.14	+2.20	N/A	N/A	N/A	-293	+612
(47) Regional (52 neighbors)		+1.12	N/A	-2.95	+1.30	N/A	N/A	N/A	-209	+396

Nomenclature: See separate page